Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/11/2011
PI Name:	Carleton, Stephanie M. Ph.D.		
Project Title:	A Combinatorial Approach of Exercise and Myostatin Inhibition to Enhance Compromised Bone		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
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Zip Code:	65211-1230	**Congressional District:**	9
Comments:			
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2010 (previously 12/31/2009) per NSBRI (1/10)		
Key Personnel Changes/Previous PI:			
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POSTDOCTORAL FELLOWSHIP
Osteogenesis imperfecta (OI) is a heritable connective tissue disorder due to mutations in the type I procollagen genes. The resulting phenotype is short stature, bone deformity and numerous lifetime fractures. Oim/oim mice, the most widely used mouse model of OI, are homozygous for a null mutation in the COL1A2 gene of type I collagen and have significantly reduced femoral biomechanical strength as well as altered bone mineral composition. Heterozygous mice (+/oim) have a phenotype intermediate to oim/oim and wildtype (Wt) mice.

Bone is inherently mechanosensitive, responding and adapting to its mechanical environment. Bone formation occurs in response to high mechanical loads, often changing its geometry to strengthen the skeleton. The largest physiological

Task Description:	loads bones typically experience are from muscles with bone strength directly proportional to muscle mass. Myostatin, a transforming growth factor-B (TGF-B) super family member, is a negative regulator of skeletal muscle growth. When myostatin is missing or non-functional, the result is increased muscle growth with a concomitant increase in bone strength. Previous studies demonstrated that mice completely deficient for myostatin (mstn/mstn) have a 40% increase quadriceps muscle mass, an 11% increase in bone mineral density, and a 20% increase in radial ultimate force relative to Wt mice. When mstn/mstn mice were subject to treadmill exercise, radial ultimate force improved by an additional 17%. Exercise has also been shown to be beneficial for human patients at an increased risk for fracture (i.e. osteoporosis, osteogenesis imperfecta). Based on these previous reports, this project was designed to determine if a combination of exercise and myostatin haploinsufficiency could improve the bone strength in mice with compromised bone. The aims of this project are to determine if: 1) OI model mice are able to tolerate and respond to treadmill exercise, 2) myostatin haploinsufficiency will ameliorate the bone phenotype of OI model mice and 3) if maternal myostatin deficiency influences the bone strength in her offspring. The long term goal of this project is to determine if a combination of these two therapies may be additive, further improving the bone phenotype beyond either therapy alone. This combination would be represent a novel treatment approach for OI patients, astronauts returning from microgravity, bed rest patients and others with compromised bone. Preliminary data demonstrates the reduction in bone strength seen in female +/oim mice was partially rescued when those mice either underwent treadmill exercise or were also deficient for the myostatin protein (+/mstn). However, maternal myostatin deficiency appears to impact the bone strength of offspring, regardless of gentoype. Wt mice born to myostatin deficient (+/mstn) mothers had marginally stronger bones than Wt mice born to Wt mothers. Taken together, these data indicate that both treadmill exercise and myostatin deficiency have the potential to improve bone strength in +/oim mice with compromised bone. In order to complete this study, additional female +/oim mice will be evaluated to assess the impact of myostatin haploinsufficiency on both muscle size and bone strength and stiffness. Additionally, the same studies will be performed on male +/oim mice to determine if a gender difference exists in the ability of the bone to respond to myostatin deficiency. Male and female G610C OI mice carrying a more common OI-causing mutation will also be evaluated for their ability to respond to treadmill exercise. The impact of maternal myostatin deficiency on the muscle size and bone strength of the offspring will also continue to be evaluated. Finally, +/oim mice of both genders that are also deficient for myostatin will be subjected to treadmill exercise to determine if exercise coupled with myostatin haploinsufficiency further improves bone strength in +/oim mice.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Muscle atrophy and bone loss have both been associated with the weightlessness experience by astronauts, but these are also widespread health concerns on Earth. As a person ages, muscle use tends to decrease. With disuse, muscle atrophy increases and bone strength decreases with a concomitant increase in fracture risk. While it has long been known that exercise decreases muscle atrophy and fracture risk while increasing bone strength, the role that myostatin inhibition may play in further augmenting the positive physiological effects of exercise is not known. This study was designed to address 1) the effects of exercise on the bone strength in both normal mice and mice with compromised bone whose clinical outcome mimics that seen in osteoporosis patients, 2) what impact, if any, myostatin haploinsufficiency has on bone strength, muscle physiology and muscle strength in both normal mice and mice with compromised bone, and 3) determine if maternal myostatin haploinsufficiency plays a role in determining the bone strength of the offspring. Insights into how myostatin deficiency influences bone size and strength may expose new avenues for improving bone strength in diverse populations such as patients suffering from bone loss diseases or long term immobilization as well as astronauts exposed to microgravity. The preliminary finding that myostatin deficiency improves femoral biomechanical integrity despite reductions in collagen content is intriguing and may indicate that alternative pathways or proteins are involved in the determination of bone strength. These pathways or proteins, either alone or in combination with physical activity, are potential therapeutic targets which may help astronauts and Earth bone patients ameliorate often debilitating bone loss. At the completion of the study, we will have determined whether a pharmacological approach of myostatin inhibition should be pursued, along with the inclusion of exercise, to potentially ameliorate the muscle atrophy and bone loss often seen as a consequence of aging and/or a sedentary lifestyle.
Task Progress:	Data collected thus far demonstrates the reduction in bone strength seen in female +/oim animals was partially rescued when those mice either underwent treadmill exercise or were also deficient for the myostatin protein (+/mstn). Modest improvements were seen in femoral torsional ultimate strength and energy to failure in both Wt and +/oim mice, though these improvements were not significant. Treadmill exercise did significantly reduce whole bone and material stiffness in both Wt and +/oim mice. Curiously, treadmill exercise did not appear to be beneficial for male Wt or +/oim mice. Myostatin deficiency marginally improved femoral torsional ultimate strength and energy to failure while reducing stiffness in +/oim mice, though not significantly. Myostatin deficiency also imposed muscle damage as seen by vacuoles and centralized nuclei. Taken together, these data indicate that both treadmill exercise and myostatin deficiency have the potential to improve bone strength and reduce stiffness in +/oim mice with compromised bone. However, female Wt femora had drastically different breaking strengths depending on the myostatin status of their parents. Femora from Wt females born to myostatin deficient mothers tended to be stronger than femora from Wt females born to myotatin deficient fathers. Additionally, femora from Wt females born to myostatin deficient mothers had decreased hydroxyproline, a measure of the amount of collagen, with an increase in polar moment of area, torsional ultimate strength and energy to failure. Tensile strength, a measure of the strength of the bone material, was decreased when Wt females were born to myostatin deficient fathers. At the ultrastructural level, if either or both parent was myostatin deficient, femora from Wt females had reduced phosphate/amide II and amide I/amide II ratios as seen by Fourier Transformed Infrared Spectroscopy (FTIR), suggesting differences in the conformation of the proteins. In order to complete this study, femora from three additional female +/oim mice subjected to treadmill exercise will be evaluated to assess the impact of treadmill exercise on bone strength and stiffness. In addition, analyses are ongoing to determine if myostatin deficiency improves bone strength while reducing stiffness in +/oim female mice. We are also collecting bones from Wt female offspring at 1-3 days and four months of age to be used for RT-PCR to determine if parental myostatin deficiency impacts the transcription of type I collagen genes, which may help explain the differences we see in bone strength in female Wt offspring from myostatin deficient parents. Finally, femora from male and female G610C OI mice are also being evaluated for their ability to respond to treadmill exercise to determine how a specific

	mutation influences the ability of the bone to respond to loading.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
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